

Deperias Kerre, PhD

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in Deperias Kerre

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Personal Profile

- Deperias Kerre is a researcher in artificial intelligence and semantic data management, with a particular focus on metadata quality, information retrieval, knowledge representation, neural-symbolic AI, and semantic data integration. He is currently a Postdoctoral Researcher at the Montefiore Institute of Electrical Engineering and Computer Science, University of Liège, Belgium, where he contributes to the Context-Aware Data Integration project. He also serves as a Lecturer at the School of Computing and Engineering Sciences, Strathmore University, Kenya.
- Deperias holds a BSc in Mathematics and Computer Science (First Class Honours) from Kirinyaga University in 2018, an MSc in Information Technology (Artificial Intelligence and Data Analytics) from Strathmore University in 2021, and a PhD in Computer Science from the University of Montpellier, France in 2025. His doctoral research focused on developing information extraction and semantic data integration techniques for knowledge representation and scientific data mining, with applications in the quantum cascade laser domain.
- His long-term research goal is to improve the discoverability, interoperability, and reuse of data through semantic technologies, metadata quality assessment, and intelligent information retrieval methods. He is particularly interested in understanding how metadata quality influences data discovery and in developing techniques that enable more effective access to data in heterogeneous environments.

Education

- 2022-2025** ■ **Ph.D in Computer Science (Knowledge Based AI)**, University of Montpellier, Montpellier, France.
Thesis title: *Mining Scientific Data for the Study of the Heterostructural Design and the Optoelectronic Properties of Quantum Cascade Lasers*.

Thesis Advisors: *Prof. Anne Laurent (LIRMM, CNRS, University of Montpellier)*, *Dr. Kenneth Maussang (IES-University of Montpellier)* and *Dr. Dickson Owuor (SCES, Strathmore University)*.
Jury Members: *Prof. Sadok Ben Yahia, Prof. Cécile Favre, Dr. Marzieh Derakhshannia, Prof. Antoine Moreau, Prof. Eric Tournié*.
- 2019 – 2021** ■ **M.Sc. Information Technology (Artificial Intelligence and Data Analytics)**, Strathmore University, Nairobi, Kenya.
Thesis title: *A deep normalized neural network model for strawberry fungal leaf disease detection*.
Thesis Advisor: *Dr. Henry Muchiri, SCES, Strathmore University*.
Major Coursework: Advanced Databases, Machine Learning, Natural Language Processing, Information Systems Management, Data Analytics and Visualization, Data Science, Information Systems Security.
- 2014 – 2018** ■ **Bachelor of Science (B.Sc) Mathematics and Computer Science (First Class Honors)**, Kirinyaga University, Kerugoya, Kenya.
Major Coursework: Computer Programming, Computer Networks, Databases, Artificial Intelligence, Mathematical Statistics, Web and Mobile Application Development.

Professional Certifications

- **Cisco Certified IT Essentials Instructor**, AFRALTI (African Advanced Level Telecommunications Institute).

Skills

Programming Skills	■ High-level: C, C++, R, Java, Python
	■ Markdown: HTML, CSS
	■ Scripting: JavaScript, NodeJS
	■ Ontology Engineering, Knowledge Graphs, Graph and NoSQL Databases, Relational Databases: Cypher, OWL, RDF, RDF*, Turtle, SPARQL, SQL, NoSQL, PostgreSQL.
	■ Typesetting: Latex
	■ Data: JSON, XML, CSV
Languages	■ Luhya (Native), Swahili (Fluent), Strong reading, writing and speaking competencies for English, French (Basic).
Others	■ Organ Playing, Choir and church music trainer, Academic mentorship and Motivational speaker.

Work Experience

March 2026 to date	■ Lecturer , School of Computing and Engineering Sciences, Strathmore University. Teaching and Research position. Undergraduate and graduate mentorship/supervision.
March 2026 to February 2027	■ Postdoctoral Researcher , Montefiore Institute of Electrical Engineering and Computer Science, University of Liege, Belgium. Research on the <i>context aware data integration project (data governance domain)</i> , funded by the Fonds de la Recherche Scientifique – FNRS, in the Data Representation and Engineering (DRE) research group under the supervision of Prof. Christophe Debruyne.
January 2024 to February 2026	■ Doctoral Fellow , School of Computing and Engineering Sciences, Strathmore University. Minimal teaching hours, doctoral research and student mentorship role.
January 2022 – December 2023	■ Tutorial Fellow , School of Computing and Engineering Sciences, Strathmore University. A full teaching, research and student mentorship including curriculum development roles.
September 2019 – December 2021	■ Graduate Assistant , School of Computing and Engineering Sciences, Strathmore University. Support teaching, research, curriculum development, student mentorship and other roles as assigned by the Head of Department.
June 2019 – August 2019	■ ICT Support Assistant , ICT department, Kenya Revenue Authority, North Rift Region (Eldoret). General IT support roles including networking and machine maintenance /configuration.

Teaching Experience

Certificate Programs	■ Computer software applications.
Diploma Programs	■ Database Systems, Project Supervision, Web Application Development, Introduction to Computer Programming (with C++).
Degree Programs	■ Project Supervision, Fundamentals of Computers, Information Systems in Financial Services, IT Fundamentals, Web Application Development, Advanced Database Systems, Database Systems, Introduction to Computer Programming (with C++), Artificial Intelligence (AI) and Machine Learning.

Administrative Roles

January 2021-December 2023	■ Coordinator, Work Based Learning (Industrial Attachment), School of Computing and Engineering Sciences, Strathmore University. In charge of Sourcing, placement and evaluation of students' industrial attachment for fulltime programs in the faculty.
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Scholarships, Grants and Awards

Scholarships	■ June 2023: France Excellence PhD Scholarship (French Government-Ministry of Foreign Affairs, through the French Embassy in Kenya) to pursue a Ph.D. in Computer Science at the LIRMM (Laboratory of Computer Science, Robotics and Microelectronics of Montpellier), University of Montpellier, France.
	■ September 2019: Strathmore University Staff Development Scholarship, to pursue an M.Sc. degree in Information Technology at Strathmore University, School of Computing and Engineering Sciences, Nairobi, Kenya.
	■ September 2014: Kenyan Government scholarship to pursue a B.Sc. Degree in Mathematics and Computer Science at Kirinyaga University, Kerugoya, Kenya.
Grants	■ May 2023: Climate Change AI Innovation Grants, Future Earth. Amount: 139,500 USD. Description: Grant to support machine learning techniques for climate change, to curb illegal logging.
	■ June 2022: DSCA Afrique, CNRS Dispositif de Soutien à la Coopération avec l'Afrique sub-saharienne'. Amount: 20,000 Euros. Description: Used to support an existing research collaboration between Strathmore University (Kenya) and the University of Montpellier (France) in the form of individual exchanges between researchers working in SCES (Kenya) and LIRMM (France), research workshops, conference and publication fees.
	■ March 2022: Artificial Intelligence for Development Africa Programme. Funding Body: Canada's International Development Research Centre (IDRC) and the Swedish International Development Agency (SIDA). Call ID: IDRC-SIDA-RUFORUM/WASCAL/A2063-IRG/2022. Amount: 20,000 USD.

Projects

July 2023 – July 2024	■ Climate Change AI Research Project: PhD Research Fellow working on Implementation of Neural Network Models for Detecting Illegal Curbing by utilizing sound data. A collaboration between Strathmore University and the University of Neuchâtel.
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Projects (continued)

June 2022 – July 2024

- **DSCA Afrique, CNRS Support System for Cooperation with Africa sub-Saharan:** We explore the use of large language models and other rule-based methods in extracting information from text for designing ontology models and Knowledge graphs for knowledge representation for semiconductor lasers, with the main aim of optimizing the design process with target properties. In collaboration with researchers at LIRMM, IES and Strathmore University.

Professional Service

Conferences/Workshops

- **CARI 2026 (African Conference on Research in Computer Science and Applied Mathematics):** Program Committee (PC) Member, Computer Science Track.
- **Deep Learning Indaba 2026:** Program Committee (PC) Member, Research Track.

Conferences, Workshops and Summer Schools

Conferences

- **June 1-5, 2025:** Extended Semantic Web Conference (ESWC 2025), Portoroz, Slovenia.
- **November 9-10, 2023:** The French-speaking meetings on fuzzy logic and its applications, LFA 2023 in Bourges, France.
- **September 4-8, 2023:** 27th European Conference on Advances in Databases and Information Systems, Barcelona, Spain.
- **March 11-13, 2022:** 7th International Conference on Machine Learning Technologies (ICMLT 2022), Rome, Italy.

Workshops

- **June 2, 2025:** LLM-TEXT2KG 2025: 4th International Workshop on LLM-Integrated Knowledge Graph Generation from Text (Text2KG) Co-located with the Extended Semantic Web Conference (ESWC 2025) in Portoroz, Slovenia.
- **September 4th, 2023:** 4th International Workshop on Intelligent Data – From Data to Knowledge, Universitat Politècnica de Catalunya, Barcelona, Spain.

Summer Schools

- **June 8-14, 2025:** International Semantic Web Research Summer School (ISWS 2025), Bertinoro Italy.

Research Publications

Journal Articles

1. Kerre, D., Laurent, A., Maussang, K., & Owuor, D. (2025). A Concise Ontological Model of the Design and Optoelectronic Properties in the Quantum Cascade Laser Domain. *Semantic Web*, 16 (4), 22104968251359870. <https://doi.org/10.1177/22104968251359870>
2. Kerre, D., Laurent, A., Maussang, K., & Owuor, D. (2025). An Instruction Dataset for Extraction of the Quantum Cascade Laser Properties from Text. *Data in Brief*, 58, 111255. <https://doi.org/10.1016/j.dib.2024.111255>

Conference and Workshop Papers

1. Kerre, D., Laurent, A., Maussang, K., & Owuor, D. (2025). Semantic Enrichment of the Quantum Cascade Laser Properties in Text-A Knowledge Graph Generation Approach. In *LLM-TEXT2KG 2025*:

- 4th International Workshop on LLM-Integrated Knowledge Graph Generation from Text (TextzKG)
Co-located with the Extended Semantic Web Conference (ESWC 2025), June 1 - June 5, 2025, Portoroz,
Slovenia. pp.68-89. https://ceur-ws.org/Vol-4020/Paper_ID_4.pdf
2. Kerre, D., Laurent, A., Maussang, K., & Owuor, D. (2023). A text mining pipeline for mining the quantum cascade laser properties. In European Conference on Advances in Databases and Information Systems (pp. 393-406). Cham: Springer Nature Switzerland.
https://doi.org/10.1007/978-3-031-42941-5_34
 3. Kerre, D & Muchiri, H (2022). Detecting the Simultaneous Occurrence of Strawberry Fungal Leaf Diseases with a Deep Normalized CNN. In 2022 7th International Conference on Machine Learning Technologies, (ACM), 147-154. <https://doi.org/10.1145/3529399.3529424>

Referees

1. Prof. Anne Laurent, Professor, University of Montpellier, LIRMM (UMR CNRS 5506) - Polytech Montpellier (France). Email: anne.laurent@umontpellier.fr
2. Dr. Dickson Owuor, Senior Lecturer, School of Computing and Engineering Sciences, Strathmore University, P.O BOX 58857-City Square, Nairobi-Kenya. Email: dowuor@strathmore.edu
3. Prof. Christophe Debruyne, Assistant Professor, Montefiore Institute of Electrical Engineering and Computer Science, University of Liège, B28, Liège Belgium. Email: C.Debruyne@uliege.be.